

Your Task at a Glance

A one-read summary: what to build, what data to use, what to send us, and when
This task is compensated with **500 SR** regardless of the hiring process decision.

This sheet is the short version. The full rules and scoring are in the attached **Technical Assessment Brief**. If the two ever seem to disagree, ask us.

1. The task in plain words

Imagine a drone has flown over a green field and taken photos from above. Some parts of the field are not getting enough water. **Your job is to build an AI model that looks at an aerial photo and marks the areas that look under-irrigated.**

In practice you will:

1. Find a public dataset of aerial or drone images of fields (we do not give you data; see Section 2).
2. Build a very simple first attempt (a baseline), so there is something to compare against.
3. Train a better AI model that beats that baseline.
4. Measure honestly how good it is, on images the model never saw during training.
5. Show us where it fails and explain why.
6. Produce pictures with the dry areas highlighted, and the percentage of the image affected.

You do **not** need to fly a drone, collect your own images, build an app, or deploy anything. This is about your modelling skill and your judgement.

2. Data: where it comes from

- **We do not provide the data.** Finding and choosing suitable data is part of the test.
- Use any **public** aerial, drone or satellite dataset of farmland or vegetation whose licence allows this kind of use. One possible starting point is the **Agriculture-Vision** dataset; you are free to choose another.
- It is fine to use only part of a dataset so that training fits on a free GPU (Google Colab or Kaggle). You are not expected to pay for anything.
- If the dataset has no label that says exactly "under-irrigated", choose the closest labels (for example dry or water-stressed areas) and tell us how you decided.
- In your write-up, **declare your data**: name of the dataset, link, licence, how many images you used, and how you split them into training and test sets.

3. What you must send us

#	Document / item	What it is	Format
1	Code	Everything needed to train the model and run it on a new image	Private Git repository link, or a .zip
2	README	How to install, train, and run a prediction on one new image	README.md inside the code

#	Document / item	What it is	Format
3	Technical write-up	Your approach, data declaration, results vs. baseline, failures, limits, and what you would do next	PDF, maximum 2 pages
4	Sample results	At least 5 images with dry areas highlighted, including at least 1 where the model got it wrong	PNG/JPG in a folder
5	Model weights	The trained model file, or a download link	File or link
6	Short note	Hours you spent, and where you used AI coding assistants (allowed, but you must be able to explain your code)	A few lines in the email or README

Send everything in **one email** to **[submission email]** with the subject "AI Engineer Assessment: [Your name]".

4. Timeline

Expected effort is about **8 hours in total**, spread however you like across **7 days**. Please do not spend more than 10 hours.

When	What happens	Who
Day 0: 09/22/2026	You receive this sheet and the full brief	WaqAla
Day 0 to Day 2	Read everything and send us your questions (see Section 5). Start exploring datasets.	You
Within 1 working day	We answer your questions in writing	WaqAla
Day 2 to Day 6	Build the baseline, train the model, evaluate, analyse the errors	You
Day 6 to Day 7	Write the 2-page write-up and README, check that the code runs from a clean start	You
Day 7: 10/02/2026, 23:59 Arabia Standard Time	Submission deadline	You
Within 5 working days after submission	We review and invite shortlisted candidates to a 60-minute walkthrough (15 min you present, 30 min technical discussion, 15 min your questions)	WaqAla
Within 5 working days after the walkthrough	Decision and feedback	WaqAla

Need more time for a genuine reason? Ask **before** the deadline. It will not count against you.

5. Questions to think about before you start

You do not have to answer these to us in advance. They are here to help you plan, and we are likely to discuss them in the walkthrough. If any of them reveal something unclear in the task, **email us by the end of Day 2**.

1. Which dataset will I use, and does its licence allow it?

2. What exactly will I treat as "under-irrigated" in that dataset?
3. Will I label whole image patches as dry / not dry, or mark the dry area pixel by pixel? Why?
4. How will I split training and test images so that the test is fair (for example, no images from the same field on both sides)?
5. What is my simple baseline, and which numbers will I use to compare it with my model?
6. What can I realistically finish in 8 hours, and what will I leave out on purpose?
7. What could confuse the model (shadows, bare soil, harvested rows, field edges) and how will I check?
8. How would this need to change to work in Saudi conditions: strong sun, sandy soil, date palms, turf, centre-pivot fields?

6. Who to contact

Melissa Adam, m@rizoma.sa. Asking good questions is seen as a positive. Where we have not specified something, make a sensible assumption, write it down in your write-up, and carry on.